

Grade 6 Accelerated
Unit 15
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

For questions 1-4 use the word “**MATHEMATICS**.”

1. What are the possible outcomes of choosing a letter randomly?
M, A, T, H, E, I, C, S
2. Which outcomes have more than one possibility?
M, A, T
3. How many outcomes are included in the event that a consonant is chosen?
5 – M, T, H, C, S
4. How many outcomes are included in the event that a vowel is chosen?
3 – A, E, I

5. The list of all possible outcomes for an experiment is called the

- ☐ event.
- ☒ **sample space.**
- ☐ set.
- ☐ probability.

For questions 6-8, use the standard six-sided dice shown.

6. What is the sample space?
1, 2, 3, 4, 5, 6



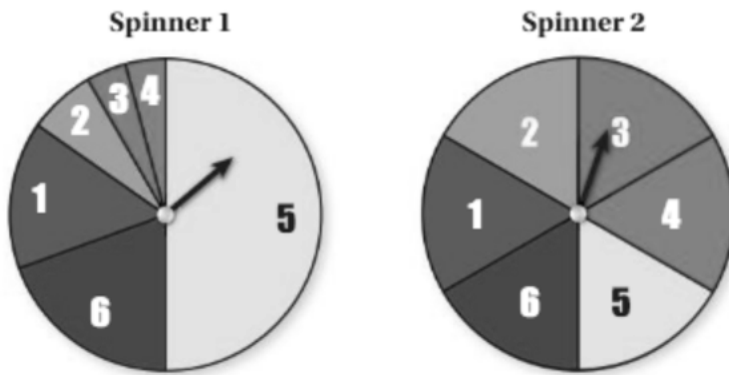
Mike and Meeka are playing a game with the six-sided dice. For each game, determine and explain whether it is fair or not.

7. Mike wins if an odd number is rolled. Meeka wins if an even number is rolled.

This game is ☒ **fair**
☐ unfair because both outcomes have an equal chance of occurring.

8. Mike wins if a number 4 or less is rolled. Meeka wins if a number 5 or greater is rolled. This game is ☐ fair ☒ unfair because Mike's outcome has a higher chance of occurring.

Use the two spinners shown below for questions 9-10.

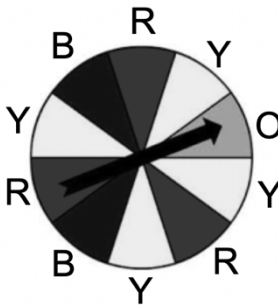


9. How many outcomes does each spinner have?
6

10. One of the spinners is fair and the other is unfair.

Spinner 1 is ☐ fair ☒ unfair because each section does not have an equal chance of being selected.

Spinner 2 is ☒ fair ☐ unfair because each section has an equal chance of being selected.

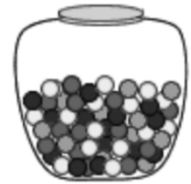
Simple Experiment	Game	Sample Space
Rolling a fair 6 sided number cube 	Player rolls a standard number cube and wins if it lands on a composite number	1, 2, 3, 4, 5, 6
Spinning the spinner 	Player spins the spinner and wins if it lands on a color that is used to make purple.	yellow, blue, red, orange

- What are the possible outcomes in the event of rolling a composite number for Game 1?
4, 6
- What fraction of the possible outcomes from Game 1 results in a win?
 $\frac{2}{6} = \frac{1}{3}$
- What are the possible outcomes in the event of spinning a color that can be used to create purple?
blue, red
- What fraction of the possible outcomes from Game 2 results in a win?
 $\frac{5}{10} = \frac{1}{2}$
- Which game does a player have a greater likelihood of winning?
Game 2

For each simple experiment, determine the sample space.

6. A glass jar contains 6 red, 4 green, 3 blue and 5 yellow marbles.

Sample Space
red, green, blue, yellow



7. The Mathletes club at school must select a day of the school week to meet for their practices.

Sample Space
Monday, Tuesday, Wednesday, Thursday, Friday

8. Drawing a face card from a standard deck of cards

Sample Space
Clubs – 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K A repeat for spades, hearts, diamonds



9. Flipping a penny

Sample Space
heads, tails

10. Selecting a random month of the year and getting a month that starts with the letter "J"?

Sample Space
January, February, March, April, May, June, July, August, September, October, November, December

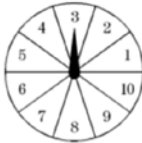
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Lesson 3 Homework

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1. Which of the following values could represent the probability of an event?

- A. -1
☒ B. 0.05
 C. $\frac{8}{6}$
 D. 225%

Determine the probability of each event occurring as a fraction, decimal, and percent. Round decimals to the nearest hundredth if necessary.

	Event	Fraction	Decimal	Percent
2.	Landing on an even number on a 12-sided die  P(even)	$\frac{6}{12} = \frac{1}{2}$	0.5	50%
3.	The spinner lands on a prime number  P(prime)	$\frac{4}{10} = \frac{2}{5}$	0.4	40%
4.	Pulling a yellow marble from a jar that has 5 blue, 4 green, and 8 red marbles. P(yellow)	$\frac{0}{17}$	0.0	0%
5.	Picking a day of the week with the letter "n" in the spelling. P(n)	$\frac{3}{7}$	0.43	43%

Use the events from the table for questions 6-7.

6. Explain which event is least likely to occur.

Pulling a yellow marble from a jar with no yellow marbles.

7. Explain which event is most likely to occur.

Landing on an even number on a 12-sided die.

Samantha is practicing her multiplication facts using the flash cards below. If she chooses a card randomly, find the probability of each even as a fraction, decimal, and percent. Round decimals to the nearest hundredth and percentages rounded to the nearest tenth.

12×3	8×3	6×6	7×4	6×8
7×5	3×12	7×6	8×5	10×6
4×9	12×4	9×6	4×6	8×4

8. $P(\text{product of } 36) = \frac{4}{15} = 0.27 = 26.7\%$

9. $P(\text{product of } 40 \text{ or more}) = \frac{6}{15} = \frac{2}{5} = 0.4 = 40\%$

10. $P(\text{product not } 48) = \frac{13}{15} = 0.87 = 86.7\%$

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Lesson 4 Homework

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1. The probabilities of four events are shown. List them in order from least to greatest.

$$\frac{4}{11}$$

5 out of 8

55%

0.46

$$\frac{4}{11}$$

0.46

55%

5 out of 8

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

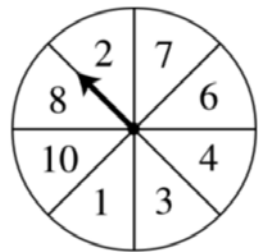
impossible

unlikely

equally likely as
unlikely

likely

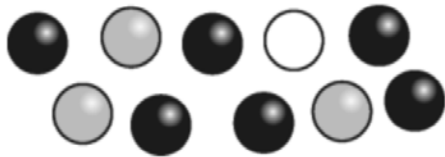
certain



2. The probability of landing on an even number, $P(\text{even})$, is 62.5%. This event is likely.
3. The probability of landing on a number less than 20, $P(\text{less than } 20)$, is 100%. This event is certain.
4. The probability of landing on prime number, $P(\text{prime})$, is 37.5%. This event is unlikely.
5. The probability of landing on a 5, $P(5)$, is 0%. This event is impossible.
6. The probability of landing on a number that is not a 1, $P(\text{not a } 1)$, is 87.5%. This event is likely.

For questions 7-10, determine which event is more likely.

7. Martin put 10 marbles in a jar. The probability he chooses a black marble is 0.6. The probability of choosing a white marble is 10%. Which is more likely to occur?



Black marble: 0.6, white marble: 0.10
Choosing a black marble is more likely.

8. Two students roll the die. Player *A* wins if they roll an even number. Player *B* wins if they roll a number less than 3. Who has the higher probability of winning? Explain.



Player *A*: $\frac{3}{6} = 0.50$

Player *B*: $\frac{2}{6} = 0.\bar{3}$

Player *A* has a greater probability, 0.50, than Player *B*, $0.\bar{3}$.

9. Fort Lauderdale airport reported that on a given day, the probability that an airline arrived on time was 0.435. The likelihood that the next airplane arrived earlier than scheduled was $\frac{1}{6}$ and the later than scheduled was 27%. What is most likely to occur when the next airplane arrives?

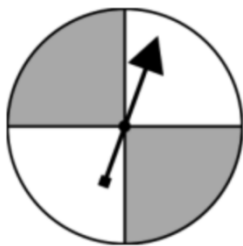
On time: 0.435

Early: $0.1\bar{6}$

Late: 0.270

The airline arriving on time is most likely to occur next.

10. The following spinner is spun. On the next spin, which color is the spinner more likely to land on? Explain.



They are equally likely because the probability is the same for both colors, $\frac{1}{2}$ or 50%.

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Lesson 5 Homework

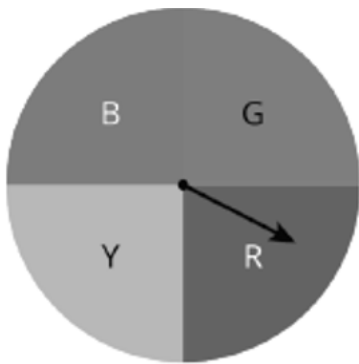
Name Answer Key
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1. Explain the difference between theoretical and experimental probability of a simple experiment.

Theoretical is $\frac{\text{outcome}}{\text{total events in sample space}}$.

Experimental is $\frac{\text{times the event occurs}}{\text{total number of trials}}$.

Tyriek performed a simulation and spun the arrow on the spinner below. He recorded his results in the frequency table below.



Color	Frequency
Blue	7
Green	9
Red	4
Yellow	5

2. What is the theoretical probability of landing on red?

$$\frac{1}{4}$$

3. What is the experimental probability of landing on red?

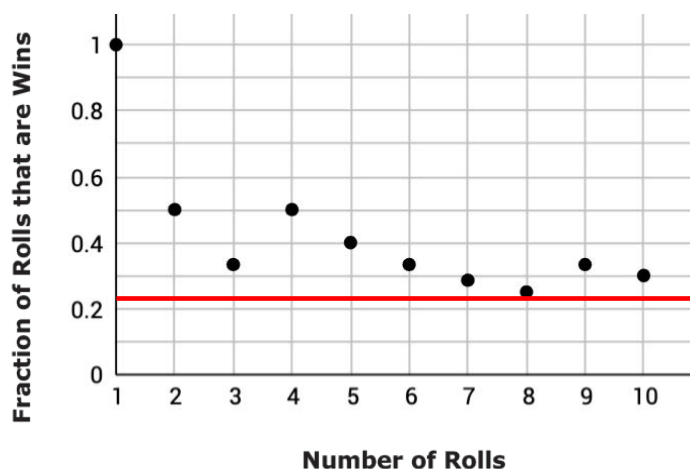
$$\frac{4}{7+9+4+5} = \frac{4}{25}$$

Arleni is playing a game where she wins if she rolls a 5 or a 6 on a standard number cube.

4. What is the theoretical probability that Alreni wins the game?

$$\frac{2}{6} \text{ or } \frac{1}{3}$$

She then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

0.3

6. What type of probability is this?

experimental

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 5 or a 6.

8. After 10 rolls, what fraction of the total rolls were a win?

$\frac{3}{10}$

9. If Arleni rolled the number cube 10 more times, estimate the probability that she will roll a 5 or a 6.

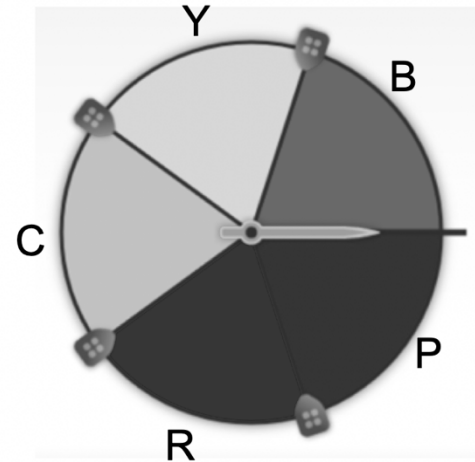
$\frac{1}{3}$

10. If she were to roll the number cube 500, how many times would you expect for her to roll a 5 or a 6

$\frac{1}{3}(500) \approx 167$ times

1. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Theoretical Probability
Cyan	$\frac{1}{5} = 20\%$
Yellow	20%
Blue	20%
Purple	20%
Red	20%



A computer simulator spinner performed more trials. The results as percentages are shown in the table below.

Color	Simulation 1 (10 trials)	Simulation 2 (100 trials)	Simulation 3 (1,000 trials)
Cyan	0%	19%	20%
Yellow	30%	23%	20%
Blue	30%	21%	20%
Purple	10%	16%	20%
Red	30%	21%	20%

2. What do you notice in comparing Simulation 1 to Simulation 2?

With more simulations, the probabilities for each color get closer to the theoretical probability.

3. Complete the table by estimating the percentages for Simulation 3.

Your teacher gives you a standard number cube with numbers 1–6 on its faces. You roll it 500 times. Your results are shown in the table below.

Outcome	1	2	3	4	5	6
Frequency	78	92	75	90	76	89

4. What is the experimental probability of rolling an even number?

$$\frac{271}{500} = 0.542 = 54.2\%$$

5. How many times would you expect to roll an even number?

$$\frac{3}{6} = 0.5 = 50\% \text{ of the time}$$

6. As the number of trials in this simple experiment increases, the

gets closer to the

- ☐ experimental probability.
- ☒ theoretical probability.

- ☒ experimental probability
- ☐ theoretical probability

Four situations are described. For each situation, create a possible simulation to fairly represent it.

Answers may vary.

7. Six people are going out to lunch together. One of them will be selected at random to choose which restaurant to go to. Who gets to choose?

Rolling a fair six-sided die to select which person will get to choose.

8. When a toddler walks, it is equally likely to step forward with its left foot or its right foot. Which foot will it use for its first step?

Flipping a coin where heads represent stepping forward with its left foot, and tail's represents stepping forward with its right foot.

9. Your school is taking 4 buses of students on a field trip. Will you be assigned to the same bus that your math teacher is riding on?

Spin a four colored spinner with one of the colors representing the bus your math teacher is riding on.

10. An experiment will produce one of ten different outcomes with equal probability for each. Why would using a standard number cube to simulate the experiment be a bad choice?

The experiment has ten possible outcomes, while a standard number cube has only six possible outcomes.

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Lesson 7 Homework

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The Smith family has 3 children and want one more child to complete their family. $\frac{1}{3}$ of the children are girls and the rest are boys.

1. What is the probability that the next child will be a girl, represented as a percent rounded to the nearest hundredth?

$$\frac{1}{2} = 0.5 = 50\%$$

2. What is the theoretical probability that the next child will be a girl?

$$\frac{1}{2}$$

Describe how each of the following can be used to simulate if the next child will be a girl.

3. A spinner with 10 equal sections.

Answers may vary.

Half of the sections represent a boy, the other half represent a girl.

4. A fair six-sided die.

Answers may vary.

Even numbers represent a boy, odd numbers represent a girl.

5. A standard 52-card deck.

Answers may vary.

Hearts and diamonds represent a boy, clubs and spades represents a girl.

The weather forecast says there is a 50% chance it will rain today.

Describe how each of the following can be used to simulate if it will rain today.

6. A coin

heads – rain	or	heads – no rain
tails – no rain		tails - rain

7. Different numbers in a box

Answers may vary.

even numbers – rain

odd numbers – no rain

*as long as there is an even amount
of numbers in the box

There are 4 types of blood: A, B, AB, and O. Blood type is determined by genes inherited from parents. Suppose that a type AB blood donor is needed at a local hospital for a patient. 0.4 donors have type AB blood.

8. Find the probability that the next donor will not have type AB blood?

0.6

9. What is the theoretical probability the next donor will have type AB blood?

$\frac{1}{4}$

10. Describe a way to simulate the probability the next donor will have type AB blood?

Answers may vary.

Spin a four colored spinner with one of the colors representing AB blood.